A Two-Stage Hierarchical Multiclass Network Intrusion Detection System

with Tempered Class Weighting and a Reject Option for Imbalanced Network Flows

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1 INTRODUCTION

Machine-learning intrusion detectors often report accuracy above 99%, yet miss the rare attack families that matter most. Two causes recur: severe class imbalance, which accuracy hides, and evaluation leakage from duplicate records crossing the train–test split. This thesis builds a leakage-safe, imbalance-aware two-stage system that detects attacks and names their family on CIC-UNSW-NB15, judged by macro-F1 rather than accuracy.

97.5% of full-release flows are benign

10,392 : 1 benign to Worms flows, after deduplication

26.4% of full-release records are duplicates

Research Objectives

- Build a leakage-safe pipeline: deduplicate first, fit every label-dependent step on training data only
- Compare class-imbalance treatments (class weighting, tempered weighting, SMOTE) on rare families
- Design a two-stage gate + family ensemble and test each component with paired bootstrap tests
- Evaluate at the realistic 97.5% benign base rate and under perturbed inputs

2 METHODOLOGY

The final system was trained on 2,085,442 flows and tested once on 521,361 held-out flows of the full release (3.54M flows).

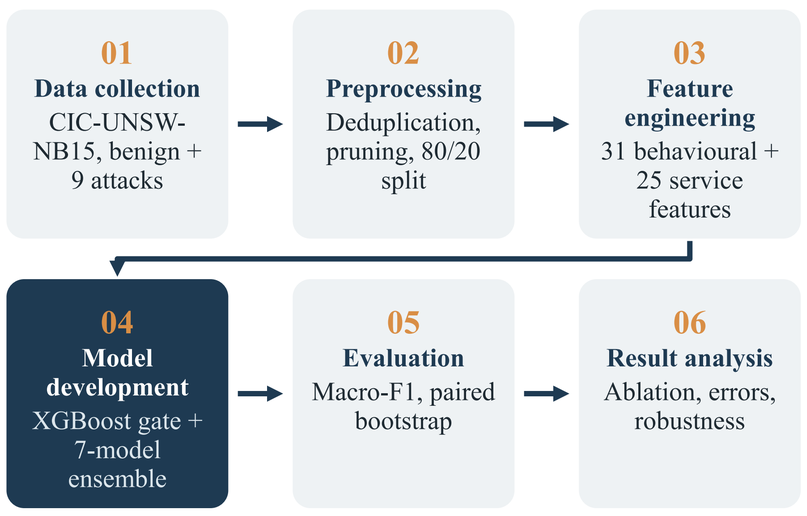


Figure 1: Research flow, from data collection to result analysis

3 DATASET & PREPROCESSING

CIC-UNSW-NB15 (2024) re-extracts UNSW-NB15 traffic with CICFlowMeter: 76 flow statistics per record. Steps that use labels run after the split, on training data only.

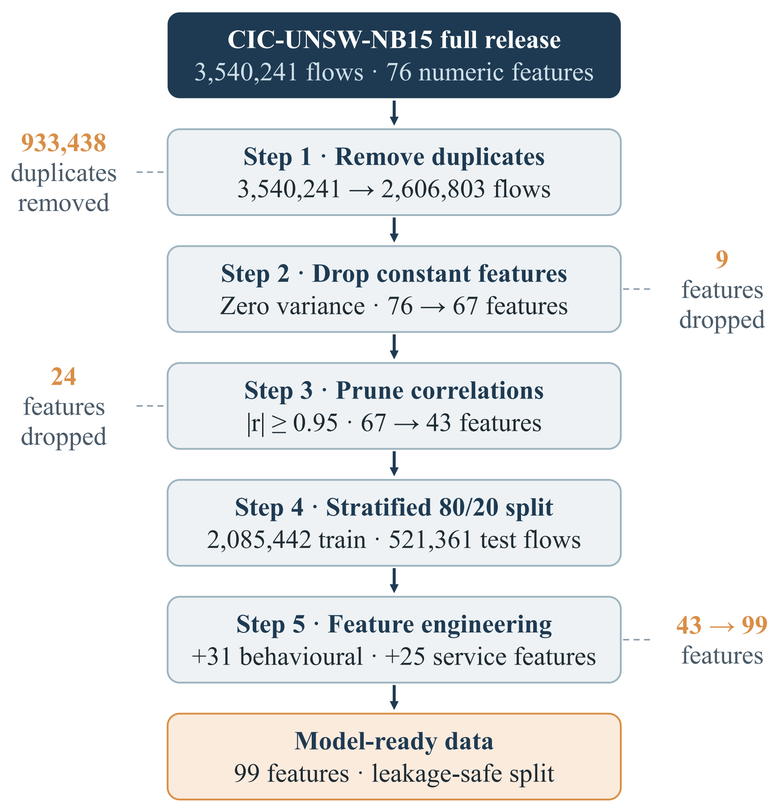


Figure 2: Preprocessing pipeline, full release

Attack flows per family, full release (after deduplication)

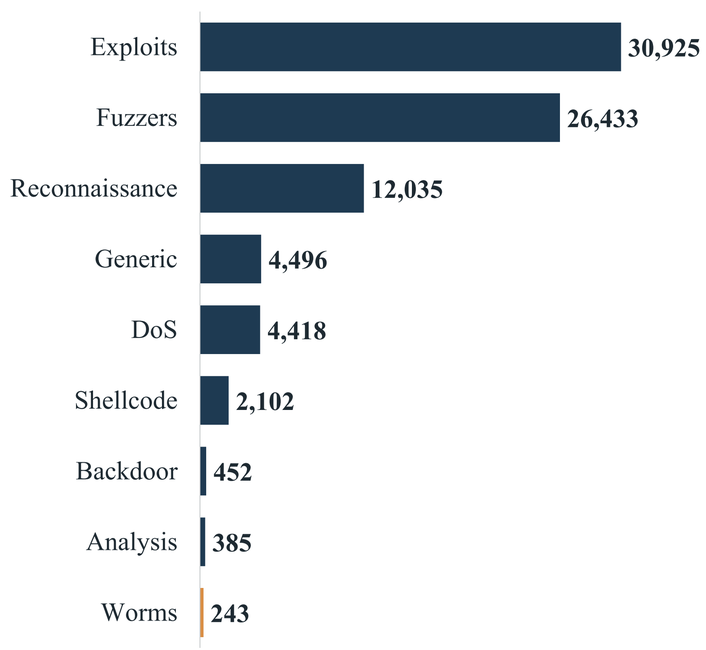


Figure 3: Class imbalance. Worms has 243 flows against 2,525,314 benign

4 SYSTEM ARCHITECTURE

A binary gate screens every flow; only suspected attacks reach a seven-member ensemble that names the family. Members differ in how strongly they rebalance classes.

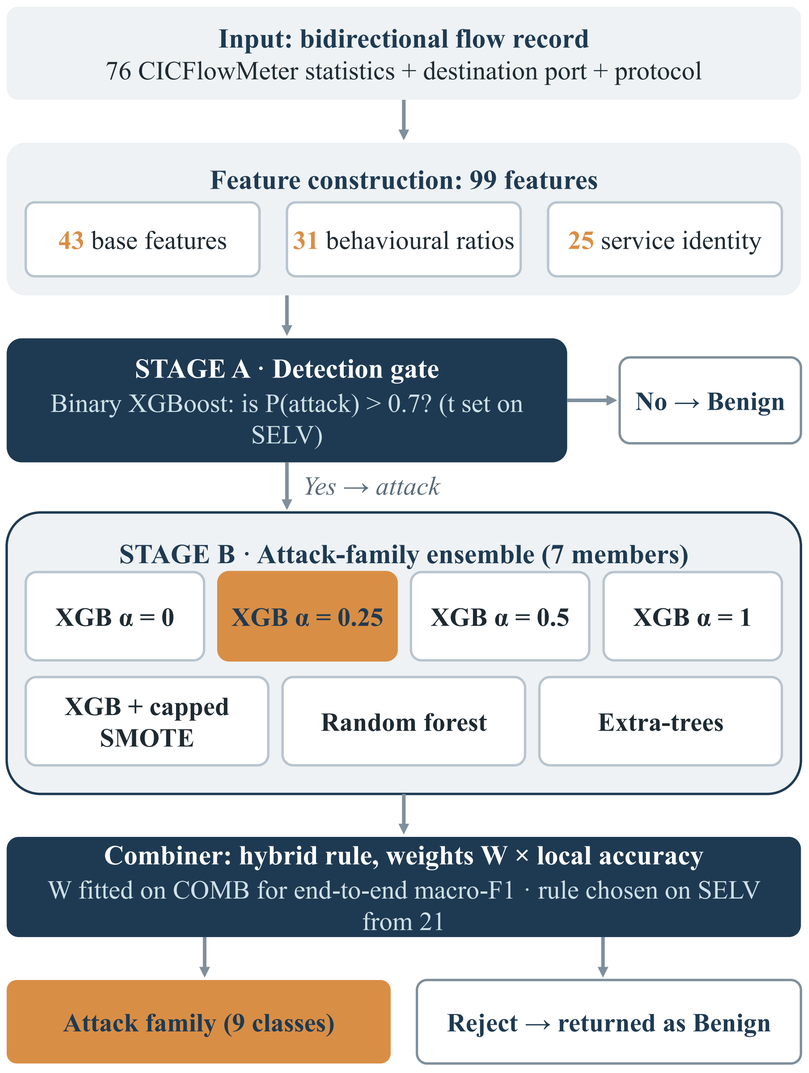


Figure 4: Architecture; COMB and SELV are the two halves of the training data

Leakage-safe Protocol

- Duplicates removed before the train–test split
- Combiner weights fitted on cross-fitted out-of-fold predictions
- Test set scored once; every gain checked with a paired bootstrap (2,000 resamples)

5 RESULTS

0.689 macro-F1, full release

98.66% accuracy, full-release test set

2.6× fewer benign false alarms

Macro-F1 by configuration, full-release test set (521,361 flows)

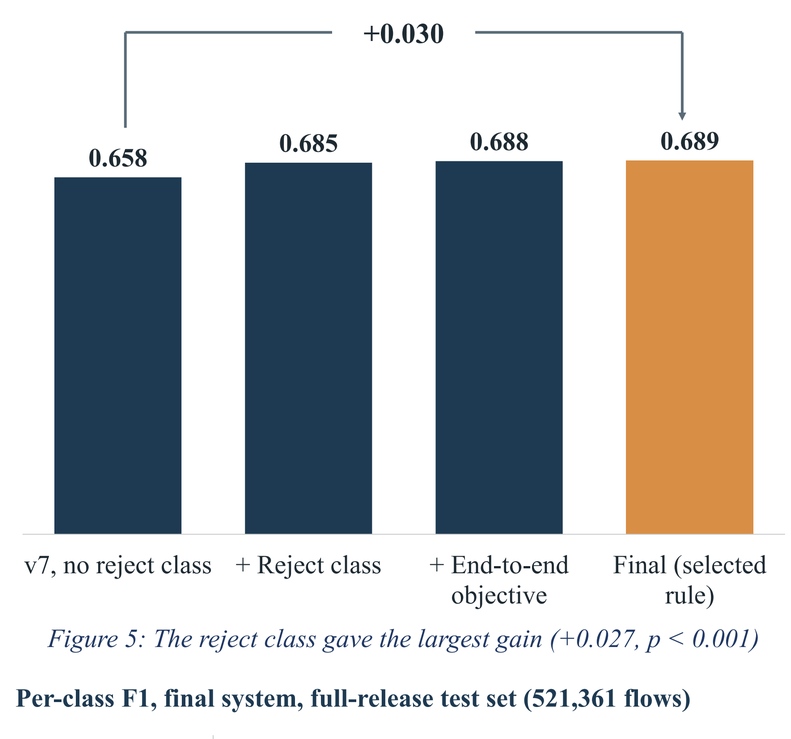


Figure 5: The reject class gave the largest gain (+0.027, $p < 0.001$)

Per-class F1, final system, full-release test set (521,361 flows)

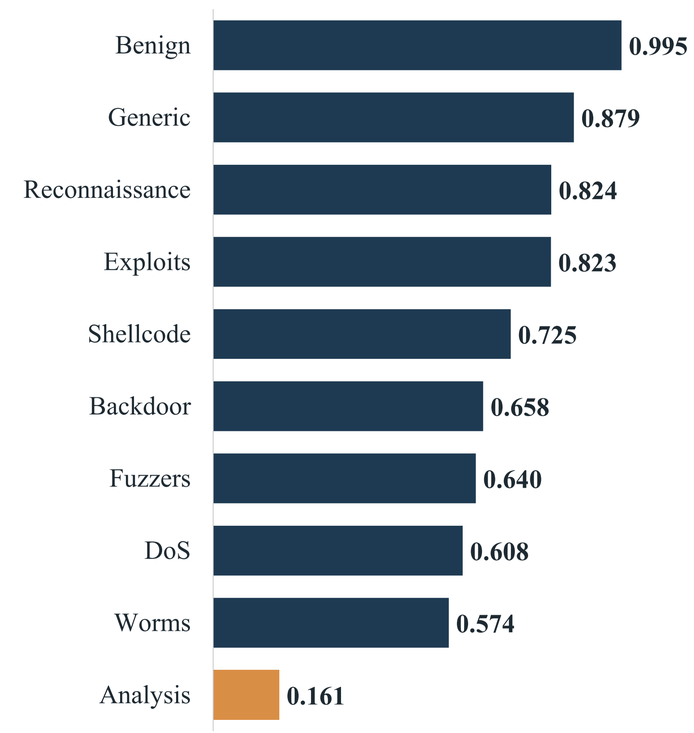


Figure 6: Analysis is the weakest family; 47 of its 77 test flows were labelled benign

5 RESULTS (CONTINUED)

Results on Other Datasets

Dataset	Macro-F1	Accuracy	Key observation
CIC-UNSW-NB15 full release (this work)	0.689	0.987	Main result, 97.5% benign
UNSW-NB15, original release	0.639	0.830	Service fields dominate
CIC-IDS2017, 13 classes	0.984	0.999	Too easy: 1-NN gets 99.7%
TON_IoT, network subset	0.960	0.981	Source IP alone flags 99.98% of attacks

Table 1: The same pipeline run on three other datasets

- On the original UNSW-NB15 release, service fields again carried most of the predictive power, confirming the service-identity result
- High scores on CIC-IDS2017 and TON_IoT come from easy testbeds, not a better model, so they cannot serve as fair second tests

6 ANALYSIS & GENERALIZATION

Gains came from better information and moderate rebalancing, not from a bigger model.

Macro-F1 gain per component, paired bootstrap

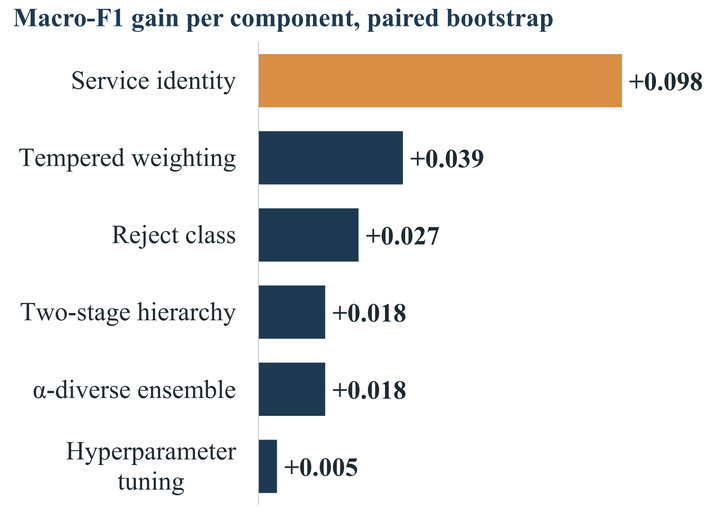


Figure 7: Hyperparameter tuning (+0.005) was not significant ($p \approx 0.23$)

- Errors come from class overlap, not label noise: only 0.35% of flows have conflicting labels
- k-NN overlap tracks model confusions (Spearman 0.827); Exploits and Fuzzers were confused 1,287 times

7 LIMITATIONS & FUTURE WORK

Limitations

- One main testbed; the other datasets are too easy
- Flow statistics only, no packet payloads
- Analysis (F1 0.161) and Worms (F1 0.574) remain weak

Future Work

- Payload features for families that flows cannot separate
- Train on one network, test on another
- SHAP explanations and adversarial training

8 CONCLUSION

- Final system: macro-F1 0.689 and 98.66% accuracy at the realistic 97.5% benign base rate
- Moderate rebalancing ($\alpha = 0.25$, +0.039) beat full SMOTE and "balanced" weights
- Information beat model capacity: service identity added +0.098; tuning gave no established gain

9 REFERENCES

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